

COMP9414 Artificial Intelligence

Tutorial topic: reasoning with uncertainty with ai9414

Example solutions

UNSW Sydney

1 Environment check

A correct local setup should include the `uncertainty` demo in `ai9414` `list`. Launching:

```
ai9414 demo uncertainty
```

or, inside the repository:

```
uv run ai9414 demo uncertainty
```

loads the default Week 7 Bayes-filter corridor demo. In the current repository, the default example is:

Item	Value
Example name	<code>two_doors</code>
Example title	Two identical doors
Cells	10
Door cells	2 and 6
Initial belief	0.1 on every cell
Sensor model	hit = 0.8, false alarm = 0.2
Motion model	success = 0.8, stay = 0.1, overshoot = 0.1

2 Normalisation

The unnormalised values are:

$$[0.04, 0.16, 0.04, 0.16, 0.04].$$

The normalising constant is:

$$Z = 0.04 + 0.16 + 0.04 + 0.16 + 0.04 = 0.44.$$

Therefore:

$$\left[\frac{0.04}{0.44}, \frac{0.16}{0.44}, \frac{0.04}{0.44}, \frac{0.16}{0.44}, \frac{0.04}{0.44} \right] = [0.091, 0.364, 0.091, 0.364, 0.091].$$

The distribution sums to one. Cells 2 and 4 are tied as the most likely cells.

3 Bayes' rule door-sensor calculation

Let D mean “the robot is at a door cell” and S mean “the sensor says door”. The tutorial gives:

$$P(D) = 0.2, \quad P(S \mid D) = 0.8, \quad P(S \mid \neg D) = 0.2.$$

Then:

$$\begin{aligned} P(D \mid S) &= \frac{0.8 \times 0.2}{0.8 \times 0.2 + 0.2 \times 0.8} \\ &= \frac{0.16}{0.32} \\ &= 0.5. \end{aligned}$$

The posterior is not 0.8 because 0.8 is the likelihood $P(S \mid D)$, not the reverse probability $P(D \mid S)$. The posterior also depends on the prior probability of being at a door cell and the false alarm rate.

If the prior changes to $P(D) = 0.05$:

$$P(D \mid S) = \frac{0.8 \times 0.05}{0.8 \times 0.05 + 0.2 \times 0.95} = \frac{0.04}{0.23} \approx 0.174.$$

If the false alarm rate rises to 0.5 while the prior is 0.2:

$$P(D \mid S) = \frac{0.8 \times 0.2}{0.8 \times 0.2 + 0.5 \times 0.8} = \frac{0.16}{0.56} \approx 0.286.$$

4 No-door sensor update

The same 5-cell corridor has doors at cells 2 and 4. The prior is uniform:

$$[0.2, 0.2, 0.2, 0.2, 0.2].$$

For a no-door observation:

$$\begin{aligned} P(\text{no door} \mid \text{door cell}) &= 1 - 0.8 = 0.2, \\ P(\text{no door} \mid \text{non-door cell}) &= 1 - 0.2 = 0.8. \end{aligned}$$

Cell	1	2	3	4	5
Prior	0.2	0.2	0.2	0.2	0.2
Likelihood	0.8	0.2	0.8	0.2	0.8
Unnormalised	0.16	0.04	0.16	0.04	0.16

$$Z = 0.16 + 0.04 + 0.16 + 0.04 + 0.16 = 0.56.$$

The posterior is:

$$[0.286, 0.071, 0.286, 0.071, 0.286].$$

Cells 1, 3, and 5 are tied as most likely. This is sensible: a no-door reading supports non-door cells.

5 Motion update

Use the exact posterior after a positive door reading:

$$\left[\frac{1}{11}, \frac{4}{11}, \frac{1}{11}, \frac{4}{11}, \frac{1}{11} \right] \approx [0.091, 0.364, 0.091, 0.364, 0.091].$$

From cell 2, the belief mass is $4/11$. Under the default right-move model:

Outcome from cell 2	Contribution	Approx. value
stay in cell 2	$(4/11) \times 0.1$	0.036
move to cell 3	$(4/11) \times 0.8$	0.291
overshoot to cell 4	$(4/11) \times 0.1$	0.036

The full predicted belief is:

$$[0.009, 0.109, 0.309, 0.145, 0.427].$$

Cell 5 receives probability mass from:

- cell 3 overshooting two cells right;
- cell 4 moving one cell right;
- cell 4 overshooting and being clamped to the last cell;
- cell 5 staying;
- cell 5 attempting to move right and being clamped to the last cell;
- cell 5 attempting to overshoot and being clamped to the last cell.

Uncertain motion usually spreads belief because one source state can lead to several destination states, and several source states can contribute to the same destination state.

6 Two-door demo trace

For the `two_doors` example, the scripted events are:

1. sense door;
2. move right;
3. sense no door.

The key belief states are:

Step	Dominant probabilities, rounded	Most likely cells
Initial	$P = 0.100$ for every cell	all cells tied
Sense door	$P(2) = P(6) = 0.250$; every other cell has $P = 0.063$	2 and 6
Move right	$P(3) = P(7) = 0.213$; $P(10) = 0.125$	3 and 7
Sense no door	$P(3) = P(7) = 0.241$; $P(10) = 0.142$	3 and 7

The first observation is ambiguous because the corridor contains two identical door cells. Seeing a door supports both cells 2 and 6. The motion update shifts both hypotheses to the right, and the later no-door reading makes the shifted non-door hypotheses more plausible than the original door positions.

7 Live Python solution

The separate file:

```
tutorials/07-reasoning_w_uncertainty/  
reasoning_with_uncertainty_solution.py
```

contains a runnable reference implementation of:

- `normalise(values)`
- `sensor_update(belief, corridor, sensor, sees_door)`
- `motion_update_right(belief, motion)`

The core implementation is:

```
def normalise(values):  
    total = sum(values)  
    if total == 0:  
        raise ValueError("Cannot normalise values that sum to zero.")  
    return [value / total for value in values]  
  
def sensor_update(belief, corridor, sensor, sees_door):  
    unnormalised = []  
    for index, probability in enumerate(belief):  
        at_door = index in corridor.door_cells  
        if sees_door:  
            likelihood = sensor.hit if at_door else sensor.false_alarm  
        else:  
            likelihood = 1 - sensor.hit if at_door else 1 - sensor.false_alarm  
        unnormalised.append(probability * likelihood)  
    return normalise(unnormalised)  
  
def motion_update_right(belief, motion):  
    next_belief = [0.0 for _ in belief]  
    last = len(belief) - 1  
    for index, probability in enumerate(belief):  
        next_belief[index] += probability * motion.stay  
        next_belief[min(index + 1, last)] += probability * motion.success  
        next_belief[min(index + 2, last)] += probability * motion.overshoot  
    return next_belief
```

The important implementation detail is that `motion_update_right` writes into a fresh list. Updating the old belief list in place would move probability mass more than once.

8 Bad sensor

In `bad_sensor`, the model has:

`hit = 0.45,      false alarm = 0.50.`

Starting from a uniform 10-cell prior and observing a door, the posterior is:

$$[0.102, 0.092, 0.102, 0.102, 0.102, 0.092, 0.102, 0.102, 0.102, 0.102].$$

Door cells 2 and 6 become *less* likely than they were before the positive door reading. That happens because the model says the sensor is slightly more likely to report a door away from a real door than at a real door.

This is a useful warning: a score or signal is not automatically evidence in the direction its name suggests. You need to know the likelihood model, and for learned systems you need calibration evidence before treating scores as probabilities.

9 Reliability bins

The bin contains 10 predictions:

$$Y, Y, N, Y, N, Y, N, Y, N, Y.$$

There are 6 correct predictions, so empirical accuracy in the bin is:

$$\frac{6}{10} = 0.6.$$

If the average confidence in the bin was 0.85, then the model is overconfident in this bin: it predicted about 85% confidence but was correct only 60% of the time.

Before trusting this conclusion, we would want more examples, independent validation data, results across other confidence bins, and checks for class imbalance or distribution shift.

10 Calibration reflection

A confidence score is not automatically a reliable probability. A classifier can be accurate but badly calibrated, for example by saying 0.99 on examples where it is much less reliable. To treat scores as probabilities, we need calibration evidence: among many predictions assigned confidence 0.8, roughly 80% should be correct. In high-cost settings, calibration is still not enough by itself; the agent also needs an action policy that accounts for consequences and can ask for more information or defer to a human.

11 Summary checklist

- The default demo example is `two_doors`.
- Normalising $[0.04, 0.16, 0.04, 0.16, 0.04]$ gives $[0.091, 0.364, 0.091, 0.364, 0.091]$.
- The door-cell posterior for the Bayes' rule task is 0.5.
- A no-door observation in the 5-cell corridor favours cells 1, 3, and 5.
- The right-move update should use a fresh destination list.
- The `bad_sensor` example makes door cells less likely after a positive door reading because the false alarm rate exceeds the hit rate.